WORKSHEET

Matrix multiplications

1		9	5		3	8	1	10	10		10	6	9	2	7	1	7	4	

Michelangelo, aged 87

Questions

Perform the following matrix multiplications, rounding to three decimal places if necessary.

1 $\begin{bmatrix} 5 & 6 \\ 2 & -4 \end{bmatrix}\begin{bmatrix} -8 & -1 \\ -2 & 3 \end{bmatrix}$

2 $\begin{bmatrix} 5 & 4 \\ 2 & 0 \end{bmatrix}\begin{bmatrix} -7 & 1 \\ -2 & -2 \end{bmatrix}$

3 $\begin{bmatrix} -0.824 & 0.329 \\ -0.295 & 0.684 \end{bmatrix}\begin{bmatrix} 2 & -7 \\ 0 & 5 \end{bmatrix}$

4 $\begin{bmatrix} 0.061 & 0.961 \\ 0.338 & 0.449 \end{bmatrix}\begin{bmatrix} -3 & -4 \\ -2 & 7 \end{bmatrix}$

5 $\begin{bmatrix} 0.744 & -0.774 \\ 0.263 & -0.463 \end{bmatrix}\begin{bmatrix} -0.003 & -0.77 \\ 0.797 & -0.736 \end{bmatrix}$

6 $\begin{bmatrix} 4 & 5 \\ -4 & 6 \end{bmatrix}\begin{bmatrix} -12 & 6 & 7 \\ 3 & -4 & 9 \end{bmatrix}$

7 $\begin{bmatrix} 1 & -2 \\ -4 & 5 \end{bmatrix}\begin{bmatrix} -12 & 12 & 6 \\ 7 & 11 & -7 \end{bmatrix}$

8 $\begin{bmatrix} -0.915 & -0.818 \\ 0.362 & 0.086 \end{bmatrix}\begin{bmatrix} 3 & -7 & 6 \\ 8 & -4 & 3 \end{bmatrix}$

9 $\begin{bmatrix} -0.232 & 1.159 \\ -0.148 & 1.963 \end{bmatrix}\begin{bmatrix} 8 & -11 & 1 \\ 4 & 6 & 2 \end{bmatrix}$

10 $\begin{bmatrix} 1.102 & 1.374 \\ 1.227 & -0.379 \end{bmatrix}\begin{bmatrix} 6 & 2 & -3 \\ -4 & -10 & -7 \end{bmatrix}$

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Solutions

A $\begin{bmatrix} 2.780 & 9.506 & 2.086 \\ 6.668 & 13.406 & 3.778 \end{bmatrix}$

E $\begin{bmatrix} -33 & 4 & 73 \\ 66 & -48 & 26 \end{bmatrix}$

G $\begin{bmatrix} -2.105 & 6.483 \\ -1.912 & 1.791 \end{bmatrix}$

I $\begin{bmatrix} -52 & 13 \\ -8 & -14 \end{bmatrix}$

L $\begin{bmatrix} 1.116 & -11.536 & -12.924 \\ 8.878 & 6.244 & -1.028 \end{bmatrix}$

M $\begin{bmatrix} -0.619 & -0.003 \\ -0.370 & 0.138 \end{bmatrix}$

N $\begin{bmatrix} -26 & -10 & 20 \\ 83 & 7 & -59 \end{bmatrix}$

R $\begin{bmatrix} -43 & -3 \\ -14 & 2 \end{bmatrix}$

S $\begin{bmatrix} -1.648 & 7.413 \\ -0.590 & 5.485 \end{bmatrix}$

T $\begin{bmatrix} -7.129 & 4.637 & -3.624 \\ 1.774 & -2.878 & 2.430 \end{bmatrix}$

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